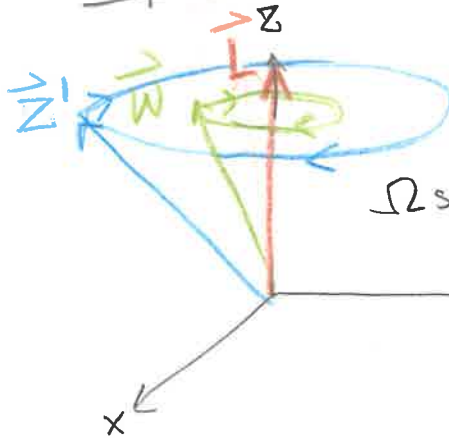


Torque-free rotation

Easier in body frame

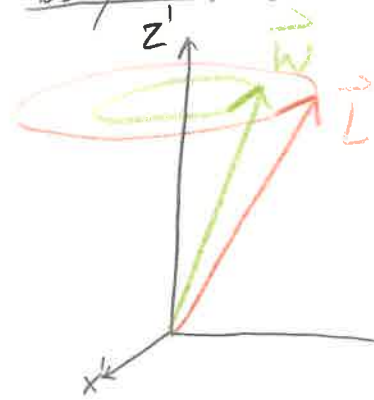
V : easier in space frame

Space frame



$$\Omega_s = \frac{1}{\lambda} \left\{ \begin{array}{l} \text{rate that} \\ \vec{S}^1 \text{ and } \vec{w} \\ \text{rotate about } \hat{z} \end{array} \right.$$

Body frame

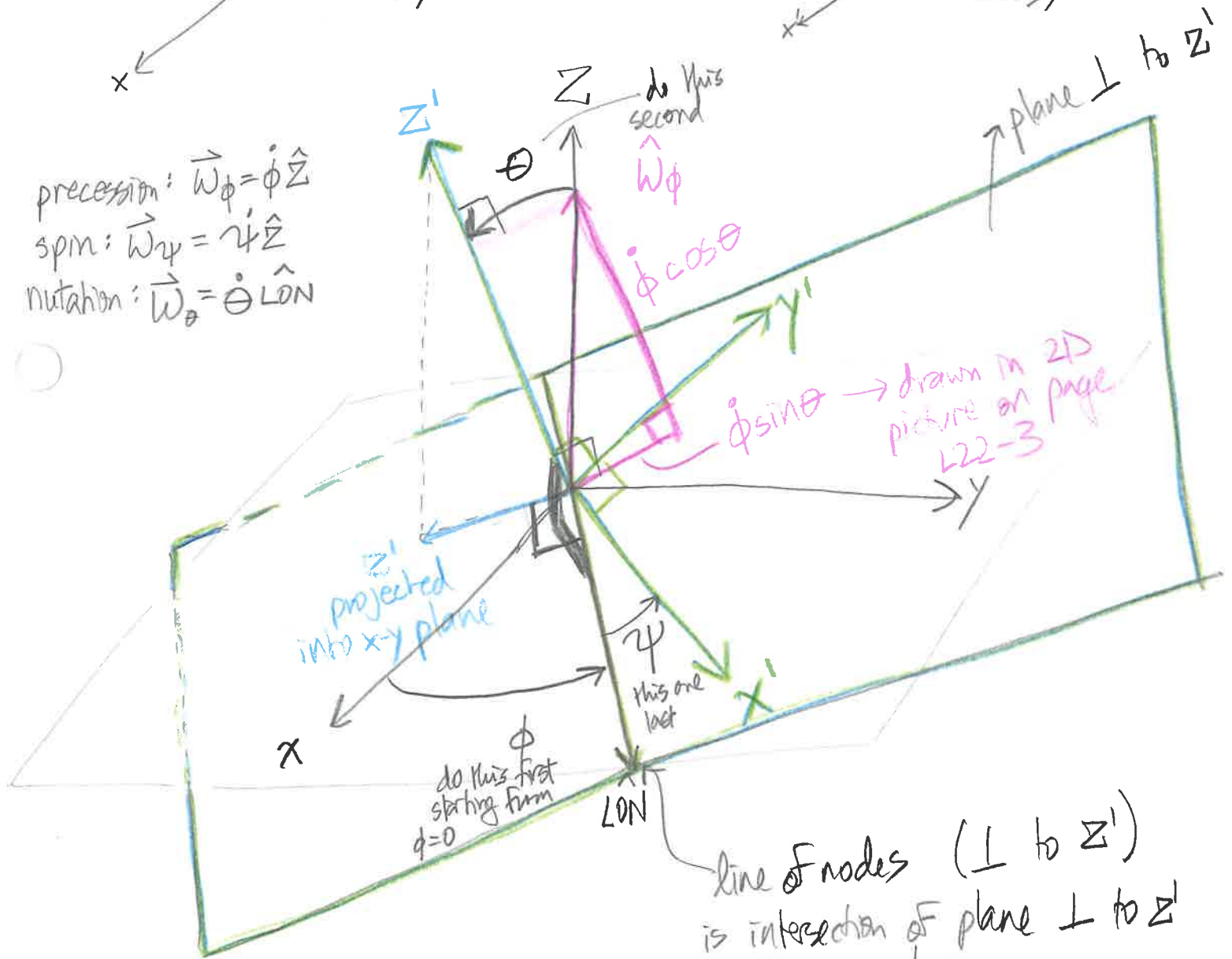


$$\Omega_b = \omega_Z' \frac{\lambda - \lambda_3}{\lambda}$$

precession: $\vec{\omega}_\phi = \dot{\phi} \hat{z}$

spin: $\vec{W}_4 = \hbar \hat{z}$

nutaton: $\vec{\omega}_\theta = \dot{\theta} \hat{LON}$



d angle from $\hat{x}$ to $\hat{LON}$

line of nodes ($\perp$ to z')
is intersection of plane $\perp$ to z'
with x - y plane

Strategy for rotating (using right hand rule)

PH III L22-2

1. Start with $\phi=0, \theta=0, \psi=0$

2. Put thumb along $\hat{z}$, spin about $\hat{z}$ by ϕ . This moves $\hat{x}$ away from $\hat{x}$.

We call this rotation $\vec{\omega}_\phi = \dot{\phi} \hat{z}$ "precession"

3. Put thumb along $\hat{L} \hat{O} \hat{N}$, rotate by θ . This moves $\hat{z}'$ away from $\hat{z}$, $\vec{\omega}_\theta = \dot{\theta} \hat{L} \hat{O} \hat{N}$

4. Put thumb along $\hat{z}'$, rotate by ψ . This moves $\hat{x}'$ away from $\hat{L} \hat{O} \hat{N}$. $\vec{\omega}_\psi = \dot{\psi} \hat{z}'$

Goal: Write down the kinetic energy T

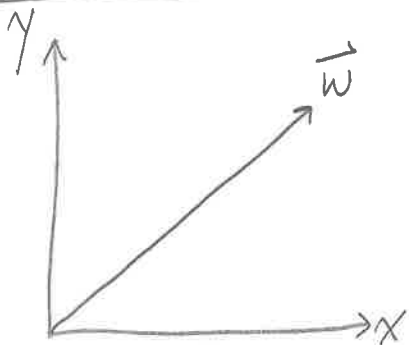
$$T = \frac{1}{2} \lambda_1 \omega_x'^2 + \frac{1}{2} \lambda_2 \omega_y'^2 + \frac{1}{2} \lambda_3 \omega_z'^2$$

We want to write this in terms of Euler angles

$$T(\phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi})$$

"painful to do but once you do it once you are done"

Infinitesimal Rotations



The rotation $\vec{\omega}$ (about $\hat{w}$) will be decomposed into rotations about $\hat{x}$ and $\hat{y}$. We can actually do this!!

Rotations about finite angles don't commute, but infinitesimal ones do.

$$\underbrace{R_y(\delta\theta)}_{\text{rotate about } y} \circ \underbrace{R_x(\delta\theta)}_{\text{rotate about } x} = R_x(\delta\theta) R_y(\delta\theta) \text{ up to order } \delta\theta^2, \text{ so as } \delta\theta \rightarrow 0 \text{ these commute}$$

For example: $\vec{\omega}_\phi = \omega_{\phi_{x'}} \hat{x}' + \omega_{\phi_{y'}} \hat{y}' + \omega_{\phi_{z'}} \hat{z}'$, similar for $\vec{\omega}_\theta, \vec{\omega}_\psi$

We want to collect terms and invert to write

Ph III L22-3

$$\omega_{x1} = \omega_{\phi x1} + \omega_{\theta x1} + \omega_{\psi x1}$$

- (1) What is the projection of spin $\vec{\omega}_{\psi} = \dot{\psi} \hat{z}$ into each of the primed components?

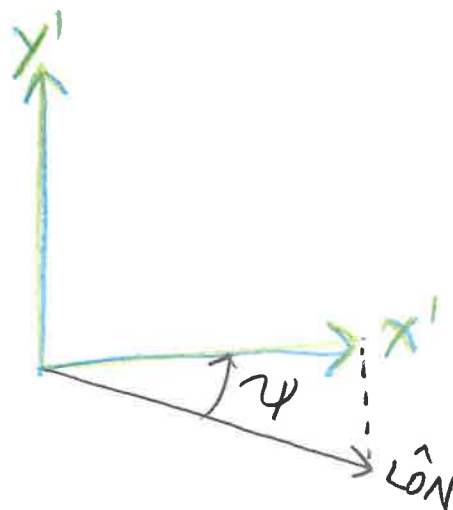
Yay! $\vec{\omega}_{\psi} = \dot{\psi} \hat{z}$, all in $\hat{z}$, none in $\hat{x}$, $\hat{y}$

- (2) What about $\vec{\omega}_{\theta} = \dot{\theta} \hat{LON}$?

$$\hat{LON} \perp \hat{z}, \text{ so } \omega_{\theta z1} = 0$$

Draw $x'-y'$ plane:

$$\hat{LON} = \cos \psi \hat{x}' - \sin \psi \hat{y}'$$



Thus

$$\vec{\omega}_{\theta} = \dot{\theta} (\cos \psi \hat{x}' - \sin \psi \hat{y}')$$

- (3) How about $\vec{\omega}_{\phi} = \dot{\phi} \hat{z}$

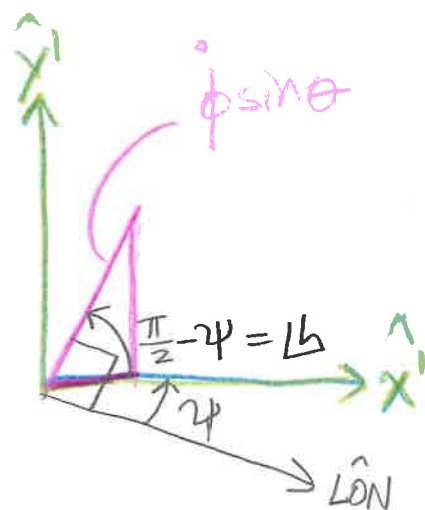
$\hat{z}'$ and $\hat{z}$ are separated by θ

$$\omega_{\phi z1} = \dot{\phi} \cos \theta$$

$$\omega_{\phi x1} = \dot{\phi} \sin \theta \cos \psi$$

$$\omega_{\phi y1} = \dot{\phi} \sin \theta \sin \psi$$

$$\vec{\omega}_{\phi} = \dot{\phi} \cos \theta \hat{z}' + \dot{\phi} \sin \theta \sin \psi \hat{x}' + \dot{\phi} \sin \theta \cos \psi \hat{y}'$$



Putting it all together: $\vec{\omega} = \vec{\omega}_\psi + \vec{\omega}_\theta + \vec{\omega}_\phi$

Ph III L22-4

$$\vec{\omega} = \begin{cases} [\dot{\theta} \cos \psi + \dot{\phi} \sin \theta \sin \psi] \hat{x}' \\ + [-\dot{\theta} \sin \psi + \dot{\phi} \sin \theta \cos \psi] \hat{y}' \\ + [\dot{\psi} + \dot{\phi} \cos \theta] \hat{z}' \end{cases}$$

Yay! We've done it!
This is useful!

We can square these components and plug into T. This is general.

Let's focus now on the simpler case of axisymmetric bodies where $\lambda_1 = \lambda_2$, separate λ_3 .

Axisymmetric bodies

$$T = \frac{1}{2} \lambda (\omega_{x'}^2 + \omega_{y'}^2) + \frac{1}{2} \lambda_3 \omega_{z'}^2$$

$$\uparrow \quad \dot{\theta} \cos \psi + \dot{\phi} \sin \theta \sin \psi$$

$$-\dot{\theta} \sin \psi + \dot{\phi} \sin \theta \cos \psi$$



$$\omega_{xy'}^2 = \omega_{x'}^2 + \omega_{y'}^2$$

but also from this diagram

$$\omega_{xy'}^2 = \dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta \quad | \text{ Plug this in}$$

(since ω_ψ is in $\hat{z}'$ direction)

$$T = \frac{1}{2} \lambda (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2$$

$$\mathcal{L} = T = \frac{1}{2} \lambda (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2$$

What is conserved?

- Hamiltonian since $\frac{\partial \mathcal{L}}{\partial t} = 0$

$H = T$ in this case, since switching to Euler angles from x, y, z , there is no explicit time dependence.

- There is a conserved momentum quantity for ψ and ϕ , since ψ and ϕ are both cyclic variables

$$p_\psi = \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) = \text{const.}$$

$$p_\phi = \lambda \sin^2 \theta \dot{\phi} + \lambda_3 (\dot{\psi} + \dot{\phi} \cos \theta) \cos \theta = \text{const.}$$

Never plug conserved quantities into $\mathcal{L}$, but we are allowed to rewrite p_ϕ in terms of p_ψ :

$$p_\phi = \lambda \sin^2 \theta \dot{\phi} + p_\psi \cos \theta$$

"Lagrangians are bad b/c they are just a crutch and don't give you intrinsic understanding"